

respatial

Design notes: project overview, mathematics, and considerations

September 18, 2026

1 Project

respatial is a research prototype for enhancing stereo recordings of multiphonic music (e.g. an orchestra). It detects the spectral activities present in a recording — no target instrument a priori — presents them to a user, and lets the user reposition each across the left/right field. The motivating case is a recording whose instruments are “crushed” to the center, which the transformation moves apart.

Three relocation modes are implemented:

- **Per-activity re-rendering** (`--pans`): NMF-based decomposition; the user names each activity and assigns a target pan; each stem is collapsed to mono and re-rendered with an equal-power pan.
- **Expand/contract** (`--stage`): a per-bin ILD warp that moves *no identified sources at all* — it exaggerates (or reduces) the per-bin left/right level difference directly, with no NMF involved.
- **Rotation** (`--rotate`): a rigid per-bin shift of every position by θ (the “rotate the stereo mic” model), which moves the whole scene without changing its spread; composes with `--stage` (rotation applied first). See the rotation notes (`respatial-rotation-notes`).

Both modes share an inverse-distance level model (Section 4.1), so that relocated content is level-adjusted as it would be on a physical stage.

2 Signal model and spatial conventions

2.1 Model

The same instrument appears in both channels with varied level but nearly identical spectrum, so in the STFT domain

$$L(t, f) \approx \sum_i a_i(t) s_i(f) e^{j\phi_i^L(t, f)}, \quad R(t, f) \approx \sum_i b_i(t) s_i(f) e^{j\phi_i^R(t, f)}. \quad (1)$$

Two consequences drive the design:

1. **Timbre is channel-invariant** $\Rightarrow$ cluster in one shared timbre space (one D for both channels).
2. **The per-bin L/R ratio is a spatial sample.** Where one activity dominates a bin, $|L/R| \approx a_i/b_i$ (its ILD) and $\angle(L/R) \approx \phi_i^L - \phi_i^R$ (its ICPD).

2.2 Conventions

Deviation from center $d \in [-90^\circ, +90^\circ]$ (each channel spans 90° ; total stage up to 180°); pan parameter $p = d/90^\circ \in [-1, 1]$. Measured ILD (dB, L/R) maps to deviation by

$$d = 2 \arctan(1/r) - 90^\circ, \quad r = 10^{\text{ild}/20}, \quad (2)$$

so +ILD (L louder) maps to left ($d < 0$). This is exactly consistent with equal-power panning: a source rendered with $g_L = \cos \varphi$, $g_R = \sin \varphi$, $\varphi = 45^\circ(1+p)$ has $r = \cot \varphi$, hence $d = 2\varphi - 90^\circ = 90^\circ p$.

3 Stage 1 — Detection

- **STFT** of both channels: Hann window, frame 4096, hop 1024 (75% overlap).
- **NMF with shared timbre**: linear magnitudes, Euclidean objective, multiplicative updates,

$$|L| \approx D A_L, \quad |R| \approx D A_R,$$

implemented by stacking $[|L|; |R|]$ and sharing one activation matrix. A fixed-seed LCG makes the decomposition reproducible between **analyze** and **remix**.

- **Spatial fingerprint** per activity: energy-gated median ILD and circular-mean ICPD over the bins assigned to it.

Naming note: what is detected are *timbral activities*, not instruments. On the real recording studied (Aranjuez guitar, 96 kHz) the NMF yields broad, overlapping spectral activities rather than clean instruments: all eight component ILDs in the 16–36 s segment fall within +1.5–+2.3 dB (deviation -10° to -15° , all slightly left).

4 Stage 2 — Per-activity re-rendering

- **Soft mask** per activity k :

$$\text{mask}_k(t, f) = \frac{D(f, k) \bar{a}_k(t)}{\sum_j D(f, j) \bar{a}_j(t)}, \quad \bar{a}_j(t) = \frac{1}{2}(A_L(t, j) + A_R(t, j)),$$

so the masks partition each bin ($\sum_k \text{mask}_k = 1$).

- **Re-render (key decision)**: each stem is collapsed to mono from the mid, $\text{mid} = (X_L + X_R)/2$, and rendered to L/R with the target pan using equal-power gains. The alternative — tweaking per-channel gains on the stereo stem — preserves original spatial detail but comb-filters when the source is moved far; a centered source has little spatial detail worth preserving, so re-rendering is preferred.
- **Residual** $(1 - \sum_k \text{mask}_k) \cdot \text{mid}$ stays centered.
- **--hard mode**: each bin is panned to its dominant component's target (no per-bin averaging; wider stage, more position jumping).

4.1 Inverse-distance gain

On a physical stage a source at center is closer to a center listener than a source at the edge, so relocated content should be level-adjusted by the inverse-distance law. With listener distance D from the stage plane, stage width W , and lateral offset $x = (W/2)(d/90^\circ)$:

$$r(d) = \sqrt{D^2 + x^2} = \frac{W}{2} \sqrt{\delta^2 + (d/90^\circ)^2}, \quad \delta = \frac{2D}{W} \text{ (“depth”).}$$

Moving content from deviation d_0 to d_1 multiplies its level by

$$g = \frac{r(d_0)}{r(d_1)}. \quad (3)$$

Content moved toward center is boosted; the center-vs-edge maximum is $20 \log_{10}(\sqrt{\delta^2 + 1}/\delta)$:

δ (depth)	1	2	5
center vs. edge	+3.0 dB	+1.0 dB	+0.2 dB

The gain is applied per component in the re-render path (using the component’s fingerprint deviation as d_0) and per bin in the warp path (Section 5). It is a real, non-negative gain, so it preserves phase. Listening verdict: $\delta = 2$ (+1 dB max) sounds most natural and is the default; $\delta = 0$ disables it.

5 Stage 3 — Expand/contract: the per-bin ILD warp

The user’s working model: position is carried mainly by the amplitude (L/R level) cue, and “expansion” means exaggerating the per-bin difference. This leads to a warp that needs no decomposition at all.

5.1 Definition

Warp the per-bin ILD by a power,

$$\text{ild}' = \gamma \text{ild} \iff r' = r^\gamma, \quad r = |L|/|R|,$$

then renormalize each bin to preserve its energy: $|L'|^2 + |R'|^2 = |L|^2 + |R|^2 = A^2$. Solving the two constraints ($|L'|/|R'| = r^\gamma$, $|L'|^2 + |R'|^2 = A^2$) gives

$$|L'| = \frac{A |L|^\gamma}{\sqrt{|L|^{2\gamma} + |R|^{2\gamma}}}, \quad |R'| = \frac{A |R|^\gamma}{\sqrt{|L|^{2\gamma} + |R|^{2\gamma}}}. \quad (4)$$

The per-channel gains $g_{L,R} = |L', R'|/|L, R|$ are real and non-negative, so *phase and ICPD are preserved exactly*; only the level ratio changes. $\gamma = 1$ is the identity; $\gamma > 1$ expands, $\gamma < 1$ contracts.

5.2 Choosing γ from a stage width

Given a target stage width S (degrees), solve by bisection for the unique γ with

$$p95(|d'|) = S/2$$

over active bins (amplitude $> 5\%$ of max), where d' is the deviation of the warped ILD. $p95(|d'|)$ is monotone in γ , so bisection on $\gamma \in [10^{-3}, 100]$ converges in < 50 iterations.

5.3 Blend

A crossfade between original and warped output,

$$\text{out} = (1 - \alpha) \text{orig} + \alpha \text{warped},$$

is also a real non-negative per-bin gain, $g = (1 - \alpha) + \alpha g_{\text{warp}}$, so it is a *pure amplitude interpolation*: phase/ICPD remain untouched and no comb-filtering can occur. α is the “amount of expansion”; combined with the stage width it gives two independent knobs (target width, and how hard each bin is pushed).

5.4 RMS matching

Per-bin energy preservation does *not* survive the overlap-add: the Hann STFT is not orthonormal, and under strong warps the time-domain energy drifts (measured -32% at $\gamma = 20$). The output is therefore scaled so its RMS matches the input’s, keeping A/B comparisons level-matched.

6 Implementation considerations

- **STFT edge artifact (fixed).** The iSTFT divides each sample by the $\sum \text{win}^2$ of the frames covering it. At the segment edges that sum tends to 0 (window tips), and for *per-bin-modified* frames (warp, masks) the unwindowed frame value near the edge does not vanish, so $y = v[k]/\text{win}[k]$ blows up: an $8\times$ peak was measured in the last 43 ms of a 20 s segment (and it clipped the 16-bit write). Fix: compute the STFT on the segment zero-padded by one frame per side; every interior sample is then covered by a frame with $\text{win} \geq 0.5$. Applied to all per-bin-modified iSTFTs (warp, stems, residual).
- **Strong-warp peaks.** Even with the edge fix, a violent warp ($\gamma = 20$) can produce time-domain peaks far above the input’s (overlap-add interference of re-weighted bins); RMS matching keeps the overall level controlled.
- **Numerics.** γ -powers are evaluated in float64 (with γ up to 100, float32 over/underflows); bins below an ε amplitude floor get gain 0 (for $\gamma < 1$ the gain diverges as $|L| \rightarrow 0$ while the product $|L|g$ stays bounded).
- **Determinism.** Fixed-seed NMF initialization so **analyze** and **remix** of the same segment give the same components.

7 Findings on a real recording

Recording: Aranjuez (guitar + orchestra), 96 kHz. Segment 16–36 s.

- **Not center-crushed.** The original’s per-bin position range is already wide: $p5..p95 = -67^\circ..+60^\circ$ (127°), $p95(|d|) = 72^\circ$. A “stage 120” target is therefore a *contraction* ($\gamma = 0.715$); a true expansion needs stage 180, which requires $\gamma \approx 20$ (the ILD distribution is concentrated near center with a long tail).
- **Full expansion ($\gamma = 20$) sounds distorted.** It pushes nearly every bin hard to one side or the other, jumping bin-to-bin (“binaural” flapping). Because the original is left-biased, most bins go hard-left and the right channel becomes sparse and spiky (measured -0.4 dB right / $+0.3$ dB left) — the perceived right-side harshness.

- **Half blend is preferred.** Stage 180 at $\alpha = 0.5$ (effective width 150°) keeps half of each bin’s original balance, has no bin-jumping, is level-matched, and was judged “crisper” — its per-bin L/R contrast is higher ($p95 |ILD|$: $16.0 \rightarrow 21.2$ dB).
- **Perceived level drop of the contraction.** Stage 120 was heard as ≈ 3 dB quieter, although measured levels are identical (L/R/mid/side/HF within ± 0.1 dB). Two mechanisms explain it:
 - *Contrast loss*: the contraction scales per-bin ILD by 0.715, dropping the per-bin L/R contrast by 2.5–3.3 dB ($p95/p99 |ILD|$) — matching the perceived 3 dB almost exactly. Reduced contrast at equal energy reads as quieter/smaller.
 - *Width→loudness*: a narrower image at equal energy sounds smaller.
- **Inverse-distance theory.** The user’s hypothesis (content moved toward center should be boosted, as it is physically closer) is correct and is implemented as `--depth`; but for this recording the mean per-bin gain is only +0.07 dB ($\delta = 2$) / +0.24 dB ($\delta = 1$) — too small to explain the perception, which the contrast mechanism does. $\delta = 2$ was judged the most natural and is the default.
- **Temporal structure.** Component activity peaks cluster in three groups (~ 3 s, ~ 8 s, ~ 16 – 18 s), but envelopes overlap heavily (pairwise cosine 0.26–0.62) — not a clean “instruments taking turns” passage.

8 Open questions / next steps

- Better decomposition: sparsity constraints, higher rank (16), envelope/transient features in the clustering.
- Smoothing the per-bin ILD (temporal/spectral median) before warping, to suppress shimmer from amplified interference.
- Per-band (frequency-dependent) panning.
- Optional loudness/contrast matching to compensate the contrast loss of contractions.
- ITD/ICPD warping: possible, but phase manipulation risks comb-filtering; amplitude-only remains the safe default.
- Minimal GUI for naming activities and assigning pans.
- Evaluation protocol: A/B listening plus per-band ILD distributions before/after.

9 References

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